

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Problem statement 16: Remote Patient Vitals Alert & Monitoring App

The system receives post-operative patient vitals, checks clinical thresholds and escalates critical alerts through a caregiver matrix. This submission focuses on collecting reliable telemetry, detecting interruptions and displaying the latest patient readings.

1. Functional requirements

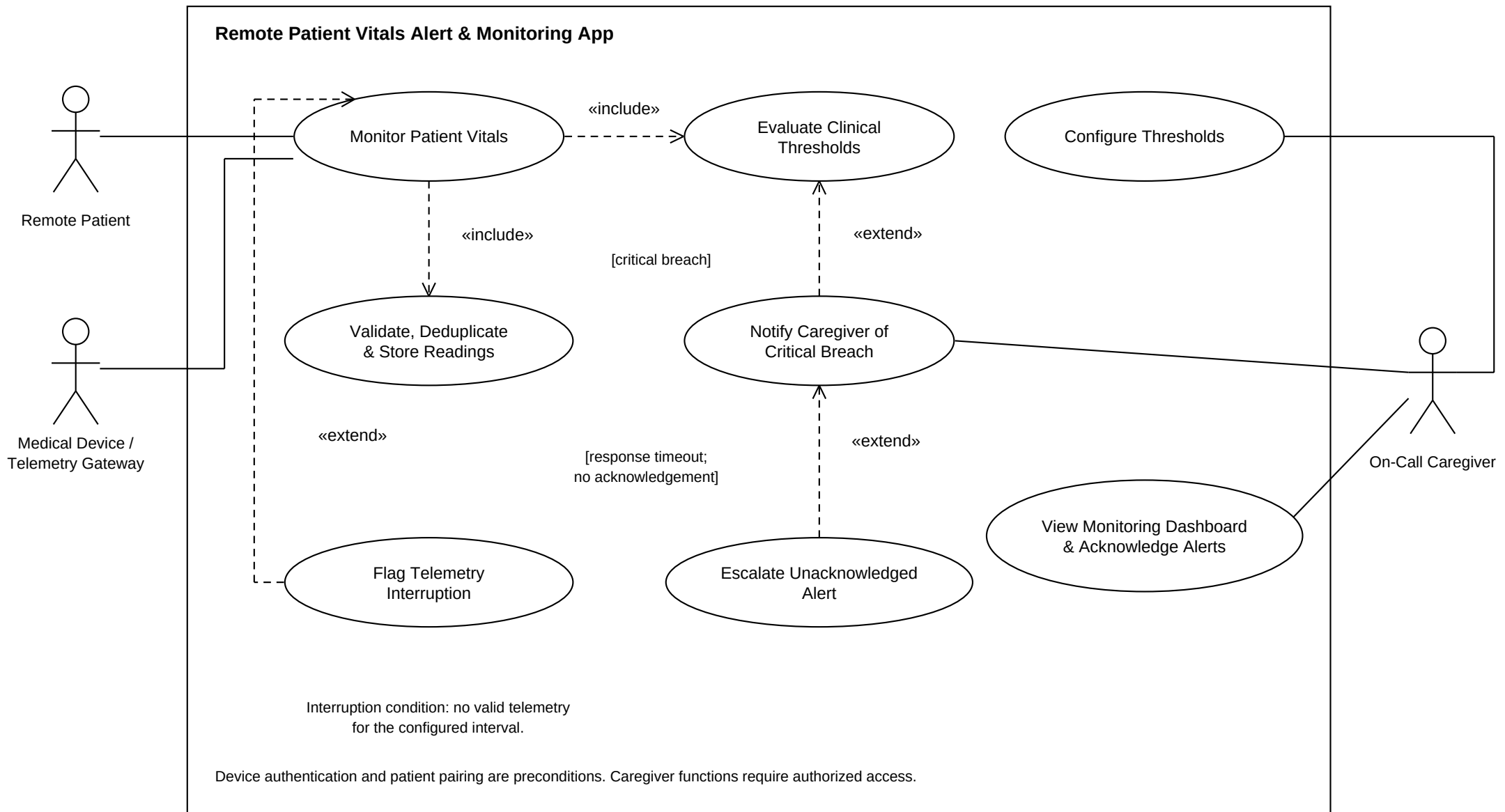
ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-001	Functional	The system shall accept SpO2, heart rate and blood pressure readings only from authenticated devices linked to a registered patient, and attach receipt timestamps.	High	Pass: A linked device's reading is associated with the correct patient and includes measurement and receipt times. An unrecognized device is rejected and its attempt logged. Fail: An unauthorized device supplies accepted data or readings are linked to the wrong patient.	Establishes a trustworthy source and patient association.
FR-002	Functional	The system shall validate telemetry fields and units, reject incomplete records and prevent duplicate storage using a unique device reading identifier.	High	Pass: A valid record is stored once. Replaying its identifier creates no second record. Missing patient identity, measurement time, required vital value or unit causes rejection with a reason. Extreme but correctly formatted values still reach clinical evaluation. Fail: Invalid or duplicate data is stored, or a critical value is discarded solely because it is abnormal.	Prevents corrupted records without hiding clinical emergencies.
FR-003	Functional	The system shall detect an interruption after the configured interval without valid telemetry, mark the stream disconnected and label the last reading as stale until valid transmission resumes.	High	Pass: With a 30-second test interval, no valid readings for 30 seconds causes Disconnected and a stale-data label. The next valid reading restores Connected and updates the display; both transitions are logged. Fail: Old data is shown as current or reconnection does not restore monitoring.	Makes gaps visible so caregivers do not rely on outdated values.

Requirements Table (continued)

ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-004	Functional	The system shall evaluate valid readings against configurable patient thresholds within 2 seconds of receipt, notify the assigned caregiver on a critical breach and escalate unanswered alerts through the caregiver matrix.	High	Pass: A 145 BPM reading with an upper limit of 140 BPM is flagged within 2 seconds. An alert identifies the patient, vital, value and time. With a 60-second response timeout, an unanswered alert is sent to the next caregiver and logged. Exhaustion is recorded if no caregiver remains. Fail: Detection is late, notification is absent or escalation follows the wrong order.	Connects reliable telemetry to the required emergency response.
FR-005	Functional	The system shall let authorized caregivers configure patient thresholds, view latest vitals with timestamps and connection status, review alert history and acknowledge alerts for assigned patients.	High	Pass: Authorized edits affect subsequent evaluation. The dashboard shows the latest stored reading and its age, status and alerts. Acknowledgement records caregiver identity and time, sets Acknowledged and stops pending escalation. Unassigned patient access is denied. Fail: Data is stale without a label, access is unauthorized or acknowledgement is not recorded.	Supports informed monitoring and accountable caregiver action.
NFR-001	Performance & availability	The telemetry gateway shall support at least 500 concurrent streams with 99.99% availability per calendar month.	High	Pass: A 60-minute test of 500 streams, each producing one valid reading per second, loses no accepted unique readings and meets the 2-second detection target. Monthly availability is at least 99.99%, measured as available time / total calendar-month time. Fail: Either target is missed.	Keeps continuous monitoring usable at the specified scale.
NFR-002	Security & auditability	The system shall encrypt patient data in transit and at rest, restrict access by authenticated identity and patient assignment, and audit device access, rejected readings, connectivity and alert events.	High	Pass: Test connections use TLS 1.2 or later; stored patient data uses AES-256 encryption. Unauthorized device and caregiver tests are denied. Each tested event records time, event type and relevant patient/device/alert identifiers; caregiver actions record identity. Fail: Data is transmitted or stored unencrypted, access checks fail or required events are missing.	Protects patient information and enables investigation of monitoring gaps.

Test settings: The stream rate, test duration and 30-/60-second timeouts are proposed verification settings, not clinical recommendations. Operational timeouts and clinical thresholds are configurable. Requirement IDs refer to this individual submission.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

UC-01: Monitor Patient Vitals

Primary actor: Medical Device / Telemetry Gateway

Other participants: Remote Patient (source of measurements); On-Call Caregiver (views monitoring status).

Trigger: A paired device begins transmitting a patient's vital readings.

Related requirements: FR-001 to FR-005; NFR-001 and NFR-002.

Preconditions

1. The patient is registered and the device is authenticated and paired to that patient.
2. The telemetry connection uses an encrypted channel.
3. Expected measurement fields, units, patient thresholds and the inactivity timeout are configured.
4. The ingestion and storage services are available.

Main success scenario

1. The device measures SpO2, heart rate and blood pressure and adds a measurement timestamp and unique reading identifier.
2. The device sends the record to the telemetry gateway through the encrypted connection.
3. The system checks device identity and patient pairing and records the receipt time.
4. The system validates required fields and units and checks that the reading identifier has not already been stored.
5. The system stores the valid, unique reading with the patient identity and both timestamps.
6. The system resets the inactivity timer and marks the telemetry stream Connected.
7. The system evaluates the reading against the patient's configured thresholds within 2 seconds of receipt. A critical breach invokes notification and, if needed, escalation under FR-004.
8. The caregiver dashboard displays the latest reading, its timestamp and age, connection status and any active alerts.
9. The system repeats the process as new readings arrive.

Alternate flow A1: Device connection is interrupted

- A1.1 After monitoring has started, no valid record arrives before the inactivity timeout.
- A1.2 The system marks the stream Disconnected and logs the interruption.
- A1.3 The dashboard retains the last reading with its timestamp and a stale-data label. Missing readings are not treated as normal readings.
- A1.4 When the device reconnects and sends a record, resume at step 3. After a valid record is stored, the system logs recovery, clears the stale label and restores Connected status.
- A1.5 If the device does not reconnect, the disconnected state and stale-data label remain visible.

Postconditions

Success: The latest valid reading is stored once, evaluated and displayed with its timestamp. The stream is Connected. Any critical breach is passed to alert handling.

Alternate outcome: During an interruption, the last reading is preserved but marked stale. Disconnection and subsequent recovery, if any, are recorded for review.